Hence acceleration along the slope decreases.

10 A track cycling competiton consists of a 4.0 km race and a one hour record race. Cyclists move around a circular track with a circumference of 1.00 km.

In the 4.0 km race, the top cyclist finishes the race in 4 mins and 1.934 s. In a one hour record race, the top cyclist travelled a distance of 55.089 km in one hour.

What is the difference between the angular speed of these top cyclists of each race?

A 0.0012 rad s^{-1} **B** 0.0077 rad s^{-1} **C** 0.014 rad s^{-1} **D** 0.92 rad s^{-1}

Ans: B

First cyclist angular speed = (4 x 2π)/(241.934) = 0.10388 rad s^{-1}

Second cyclist angular speed = (55.089 x 2π)/(60x60) = 0.09615 rad s^{-1}

Difference = 0.0077 rad s^{-1}

11 The section of a cycling track where ...